Vivek Patil

Vadodara, Gujarat
7249504782 | vivekpatilwork1125@gmail.com | linkedin.com/in/vivekpatil1125 | github.com/vivekpatil1125

Summary

I am a developer with a strong grasp of programming concepts and hands-on experience building end-to-end web applications, including real-time features and API integration. Known for combining creativity with analytical thinking to develop effective solutions. Highly adaptable, quick to learn new technologies, and motivated to grow in a dynamic professional environment.

Skills

Programming Languages: C, C++, Java, Python, JavaScript
Frontend: React.js, HTML5, CSS3
Frameworks & Backend: Node.js, Express.js, FastAPI, REST APIs
Database: SQL, MongoDB
Tools & Other: Git, GitHub, Socket.io
Generative AI / LLM & ML: Prompt Engineering, LLM APIs (Gemini, DeepSeek, Llama), AI-Assisted Development (Claude, Gemini), RAG Concepts, Basic ML Algorithms (Regression, Classification, Clustering)
Soft Skills: Communication, Teamwork, Leadership, Problem Solving, Adaptability, Quick Learner, Creative

Projects

Image Anomaly Detection Pipeline | *C++, OpenCV*

- Built a C++ computer vision pipeline using OpenCV to detect and measure internal defect regions in scanned images, modeling techniques used in industrial inspection and medical image analysis.
- Implemented a multi-stage pipeline: Gaussian blur for noise reduction, Otsu's method for automatic object segmentation, and contour detection for boundary extraction.
- Debugged a masking issue where overlapping threshold masks canceled valid regions, and resolved it by reconstructing a solid object mask via contour filling before isolating internal anomalies.
- Measured detected regions using contour area and bounding-box calculations, and rendered annotated output images for visual verification.
- Applied OOP principles and modular C++ design, separating image loading, denoising, segmentation, and measurement into distinct, reusable stages.

Code Together (Full-Stack Collaborative IDE) | *MERN, Socket.io, Monaco Editor, Judge0, GitHub API, LLMs*

- Developed a synchronized IDE with Socket.io and Monaco Editor, featuring "Pass the Mic" controls for Team and Classroom modes.
- Integrated Judge0 API for instant multi-language code execution and GitHub API for seamless in-browser commits and repo management.
- Built a parallel-fetching AI assistant (Gemini, DeepSeek, Llama) using prompt engineering techniques for side-by-side debugging, and integrated Stream.io for live video/chat.
- Implemented secure JWT auth, Zustand for state management, and optimized NoSQL schemas for social features and session history.

AI Desktop Assistant | *Java, Swing, Wit.ai, Gemini API, Apache HttpClient, Gson*

- Developed an AI-powered desktop assistant supporting both text and voice commands.
- Integrated Wit.ai for speech-to-text and Gemini API, using structured prompts for intelligent, context-aware responses.
- Implemented features like app launching, file search, system control, reminders, and web search.
- Used OOP concepts, multithreading, and event-driven programming for a modular, responsive architecture.
- Added text-to-speech via PowerShell + System.Speech and built an interactive UI using Java Swing.

PrivacyScope (Mobile Privacy Analysis App) | *Android, Permission Analysis*

- Built a mobile app that detects hidden and unreported permissions used by installed applications beyond what is shown in system settings.
- Created a system to compare declared vs. actual permission usage and alert users of privacy risks.
- Promoted data privacy awareness by exposing unauthorized access to sensitive device resources.

Education

The Maharaja Sayajirao University — Vadodara, Gujarat
Master of Computer Applications, CGPA: 75.66% — *2025 – 2027*

Sardar Patel University — Anand, Gujarat
Bachelor of Computer Applications, CGPA: 9.82 — *2022 – 2025*

Experience

Balaji College of Commerce, Science, Technology and Management
Visiting Faculty — *Aug 2025 – Jan 2026*

- Taught C Programming, HTML, CSS, and JavaScript to a class of 30 BCA students.

Achievements

10 GPA in semester 3 and semester 6, 1st rank in university (BCA semester 2, 3 and 6).

Best Project Award (BCA).

Secured 1st Rank at TechSprint 2026 – National Level Hackathon.

Secured 3rd place at Codemanthan 2.0 Hackathon.

Certifications

Software Engineering Job Simulation – JPMorganChase & Co.

IBM Web Development Fundamentals.

Advanced Course on Emerging Technologies (AI/ML, Deep Learning, Edge Computing, SAP Analytics Cloud & ABAP on BTP) – SAP Code Unnati Program & Edunet Foundation.

Languages

English | Hindi | Marathi | Gujarati